

Day 37 - NTP

Sunday, September 6, 2026

10:46 AM

NTP Info

General Information

All devices have two types of clocks, **software (clock)** and **hardware (calendar)** clocks.

- **Software** clock (aka. **Clock**) runs digitally on the operating system of the device.
- **Hardware** clock (aka. **Calendar**) is an internal physical clock built into the device.
- The default time zone for Cisco devices is **UTC (Coordinated Universal Time)**
- The most important reason to have accurate time on a device is to have accurate logs for troubleshooting. (with **show logging**)

The **calendar** of a device tracks the date and time on the device even if it restarts, power is lost, etc. When the system is restarted the **calendar** is used to initialize the **clock**. The **calendar** does have a flaw though, and it's that it will drift over time and become inaccurate, so it is not an ideal time source. That's why the Network Time Protocol (NTP) is used to mitigate this problem.

Clock and Calendar Commands

Clock/Calendar Show Commands

- **show clock**
 - You can view basic details about the **clock** through this command.
 - It'll display the current time.
 - Sometimes, an asterisk (*) will be displayed to indicate that the time is not **authoritative** (not from a trusted source)
- **show clock detail**
 - Using this command will show the previous information as well as the **time source** (set to the hardware calendar by default)

Clock/Calendar Config Commands

- **clock set HH:MM:SS DAY MONTH YEAR**
 - You can use this command to manually configure the **clock** on the device.
- **calendar set HH:MM:SS DAY MONTH YEAR**
 - You can also use this command to manually configure the **calendar** on the device.
- Typically, you will want to synchronize both the **clock** and **calendar**, which can be done through two commands:
 - **clock update-calendar** to sync the **calendar** to the **clock**'s time.
 - **clock read-calendar** to sync the **clock** to the **calendar**'s time.
- **clock timezone TIMEZONE-NAME HOURS-OFFSET [MINUTES-OFFSET]**
 - The time zone can be configured with this command.
- **clock summer-time TIMEZONE-NAME recurring START END [OFFSET]**
 - The daylight savings time (**summer time**) can be configured with this command.
 - The default offset is **60 minutes**, but it can be specified at the end if it's necessary to change it.
 - (**start** and **end** each require the week (**first**, **last** or **number**), weekday, month and time)

Network Time Protocol (NTP)

Network Time Protocol (NTP) is a protocol that allows automatic syncing of time over a network.

- It communicates over **UDP** port **123**.
- **NTP** uses the **UTC** time zone by default. The appropriate time zone must be configured on each device.
- **NTP clients** request the time from **NTP servers**.
- A device can be an **NTP client** and an **NTP server** at the same time.

Stratum Levels

Stratum Levels

Some NTP **servers** are better (more reliable) than others. That is determined through a factor which is called **stratum** level and it defines the 'distance' of an NTP **server** from the original **reference clock**. The **stratum** level range is (1 - 16), however, we start counting from (**stratum 0**).

Reference Clocks

A **reference clock** is usually a very accurate time device like an atomic clock or GPS clock. They are assigned **stratum (0)** within the NTP hierarchy.

- **NTP servers** connected directly to **reference clocks** are **stratum 1**.

NTP Hierarchy

- **Reference clocks** are **stratum (0)**.
- **Stratum (1) NTP servers** get their time from **reference clocks**.
- **Stratum (2) NTP servers** get their time from **stratum (1) NTP servers**.
- **Stratum (3) NTP servers** get their time from **stratum (2) NTP servers**.
- **Stratum (15)** is the maximum. Anything above that is considered unreliable.
- Devices can also 'peer' with devices at the same stratum to become in a **symmetric active** mode in order to provide more accurate time.

Cisco NTP operational modes

Cisco devices can operate in three NTP modes:

1. **Server** mode
2. **Client** mode
3. **Symmetric Active** mode

Stratum level names

- **NTP servers** which get their time directly from **reference clocks** are also referred to as '**primary servers**'
- **NTP servers** which get their time from other **NTP servers** are referred to as '**secondary servers**'. They operate in **server** mode and **client** mode at the same time.

NTP Configurations

NTP Show Commands

- **show ntp associations**
 - Shows connected **NTP servers**
 - It'll show you a bunch of information for each server like:
 1. The address of the **NTP servers** the device is connected to along with a symbol to indicate the status.
 2. Their **reference clocks**
 3. Their **stratum** level
- **show ntp status**
 - Shows information about the current **NTP** instance running on the device
 - It includes this info:
 1. The current device's **stratum** level.
 2. The IP of the **NTP server** which is being used as reference.

NTP Config Commands

- **ntp server IP-ADDRESS [prefer]**
 - Forms a connection to an **NTP server**.
- **ntp server IP-ADDRESS key KEY-NUMBER**
 - Forms a connection to an **NTP server** with an authentication key.
- **ntp peer IP-ADDRESS**
 - Forms a **symmetric active** connection with another **NTP** device. Both devices will have the same **stratum** level.
- **ntp peer IP-ADDRESS key KEY-NUMBER**
 - Forms a peering to another **NTP** device with an authentication key.
- **clock timezone TIMEZONE-NAME HOURS-OFFSET [MINUTES-OFFSET]**
 - Allows you to configure the time zone on the current device.
- **clock summer-time NAME recurring START END [OFFSET]**
 - Sets the daylight saving time (**summer time**) configuration for the **clock**.
 - The default offset is **60 minutes**, but it can be specified at the end if it's necessary to change it.

- (**start** and **end** each require the week (*first*, *last* or *number*), weekday, month and time)
- *ntp source INTERFACE-ID*
 - Lets you assign the interface which the **NTP** messages will source/originate from.
- *ntp master [STRATUM]*
 - This will assign the current device as an **NTP server** with an optional *stratum* level.
 - If the *stratum* level is not specified, it will default to **8** (displayed as **7** in the show commands, because *stratum* levels begin counting from **0**)
- *ntp authenticate*
 - Enables **NTP** authentication on the device.
- *ntp authentication-key KEY-NUMBER md5 KEY/PASSWORD*
 - Initializes an **NTP** key number along with a password assigned to it.
- *ntp trusted-key KEY-NUMBER*
 - Trusts an **NTP** key for it to be used in authentications.